

Formative assessment: data science mini project

Environment management and version control

This week we'll carry on working on the mini project you started last week using the NHANES data, applying your new skills in environment management and version control.

Section 1: Adding a Conda environment

You're going to be using R and the Tidyverse collection of packages to work on the mini project, so set up a new Conda environment with r-base and the Tidyverse installed.

Export the environment into a .yml file at the top level of the formative directory. Make sure you've exported it in a way that will work across different platforms.

Section 2: Adding the mini project to version control

Now you need to add the mini project to version control. However, there are some files that you don't want to add to your Git repository, either because they're temporary files or because they're not suitable for uploading to GitHub (such as data that you're licenced to use but not publish, or saved passwords or access tokens). To avoid adding these files, you'll need to create a .gitignore file at the top level of your formative directory. Have a look at the template .gitignore file linked from the GitHub lecture, and create your own that tells Git to ignore temporary files and the contents of the data directory. You'll also want to review your code at this stage to make sure it doesn't include anything that might be a security risk. For example, make sure you're using relative paths rather than absolute paths.

Now navigate to the top level of the formative directory and initialise a Git repository. Stage all the files and commit them to the repository.

Section 3: Adding a README file to the top level

We're going to add a README file to the top level formative directory. Make a new branch called "readme" and switch to it. Add a readme text file to the formative directory that includes instructions on how to create and activate the Conda environment. Stage the changes and commit them with an informative commit message.

Section 4: Pushing to GitHub

Next, we'll add the repository to GitHub for reproducibility and collaboration. Set up a new public repository on GitHub. Push your local formative repository (including the readme branch) to the GitHub repository.

Section 5:

On GitHub, create a new pull request from the readme branch to main. Then either merge the pull request yourself or, even better, ask your neighbour for their GitHub username and assign them as a reviewer. Review your neighbour's readme and either approve it or suggest changes. Then merge your own approved readme pull request into Main. Last of all, sync your local copy with the GitHub version.

Next steps

If you've finished all of this and still have time, have a look at the links in this week's Homework section on Blackboard.